

STEVEN (CHENG-JU) WU

(669) 288-4917 | cw3834@columbia.edu | linkedin.com/in/steven-cj-wu/

EDUCATION

Columbia University	New York, NY
Master of Science in Mechanical Engineering	Expected Dec 2027
Relevant Course: Robotics Studio, Continuous Control Systems, Microelectromechanical Systems, Advanced manufacturing Processes.	
Santa Clara University (SCU)	Santa Clara, CA
Bachelor of Science in Mechanical Engineering	Jun 2026
Relevant Courses: Composite Materials, Fluid Mechanics, Finite Element Theory and Applications, Engineering and Design with Applied Machine Learning, Computational Mechanics, Continuum Mechanics.	

WORK EXPERIENCE

Santa Clara University GIFTED Lab	Santa Clara, CA
Undergraduate Research Assistant	Jun 2025 - Jan 2026
- Awarded the SCU Kuehler Undergraduate Research Grant to conduct research on machine-learning-driven topology optimization. - Developed a PINN-based topology optimization framework for 60×20-element structural domains, integrating minimum feature-size constraints to generate additive-manufacturing-compatible designs. - Implemented a multi-network architecture coupling material-density prediction with displacement-field estimation, reducing optimization runtime by 5% versus SIMP-based methods in complex solutions. - Built and trained neural network models in PyTorch using automatic differentiation; analyzed and visualized material-density evolution with NumPy and Matplotlib.	
SINBON Electronics Co., Ltd	New Taipei City, TWN
R&D Mechanical Engineering Intern	Jul 2024 - Sep 2024
- Built and iterated injection-molded plastic and sheet-metal components in Creo for OEM electronics assemblies, considering manufacturability, structural requirements, tolerances, assembly interfaces, and cost. - Produced 3D CAD models, engineering drawings, BOMs, and design documentation to facilitate prototyping, supplier sourcing, assembly, and production. - Collaborated with electrical and manufacturing engineers to resolve packaging, tolerance, and DFM issues across 3 product designs, managing prototype-to-production development.	

ACTIVITIES

SCU Bronco Air Design Build Vertical Flight (DBVF)	Jun 2025 - Jun 2026
Pegasus	
- Created and fabricated an autonomous hexacopter for the Vertical Flight Society DBVF Competition, placing 9th overall. - Designed airframe and motor-arm assemblies in SOLIDWORKS, reducing structural weight by 47.4% through FEA-driven topology optimization while maintaining FOS > 2.0 and deflection < 1 mm. - Integrated Raspberry Pi and CubePilot Cube Orange+ into a 6-motor autonomous flight system, supporting waypoint navigation and mission-route planning.	
SCU Robotics System Lab	Jun 2025 - Jun 2026
Vertical Profiler	
- Diagnosed post-deployment damage on 3 vertical profilers and reverse-engineered replacement components in SOLIDWORKS for 3D printing. - Designed modular SOLIDWORKS mounts for action cameras and sensors, enabling rapid field reconfiguration across multiple instrumentation setups.	
SCU Bronco Racing	Sep 2022 - Jun 2024
Formula SAE Electric	
- Created side-impact structures in SOLIDWORKS for a Formula SAE Electric vehicle, meeting structural, cockpit-clearance, and driver-safety requirements. - Performed FEA on cockpit and side-impact structures under simulated 70 km/h side-impact and 4.5G worst-case loading, comparing material configurations for structural integrity and weight reduction.	

LANGUAGE AND IT SKILLS

- Languages: Mandarin (Native), German (Elementary).
- Technical: SOLIDWORKS, Creo, MATLAB, 3D Printing, Laser Cutting, Pytorch, Python, C++, GD&T.